

Mathematics

A Cultural Heritage

Lecture 1

Introduction: Assignment Logistics

Lecture Outline

1. Assignment logistics
2. Group Project Details
3. Group Presentation Details

Assessment

Task	Weighting
In-class worksheets	10%
Tutorial participation	20%
Written assignments	30%
Essay & presentation	40%

Assignment 1 Logistics

- Pick two articles (out of 4 possible). Choices on website
- Not a book report/summary
- Direct comparison not required, but...
 - Should have central theme
 - Should have smooth transitions/**not** just two essays “glued together”
- Describe connection to **lectures, readings, and tutorials.**
- Connect to **Course Learning Outcomes**
- 500-850 Words
- Watch out for **copying/plagiarism**
- Due date: **18th October, 2021, 16:55**

Course Learning Outcomes

1. Appreciate and describe the beauty, the utility, and the “way” of mathematics.
2. Comprehend and describe how mathematics was and is being developed as a work of human culture.
3. Investigate and describe the interplay between mathematics and other areas of human culture.
4. Investigate and explain the role of mathematics in the development of civilization.

Assignment 2 Logistics

- Start collecting clippings that have to do with mathematics from current (published on or after 15th September, 2021) **newspapers, magazines, or webpages** (in English or in Chinese, whose target audience is the general public).
- Do **not** take an excerpt out of a book or a magazine or a website which is **specifically on mathematics or science**.
- Your source should **not** be media explicitly designed for experts to write in, even if written for laypeople.
- As a **general guideline**, ask whether most people would see this media in their **daily lives** or only in a **specialized search**.
- Connection to math **explicitly** mentioned in article
- Target number of items is 10.

Assignment 2 Logistics

- Record **source** and **date**, **write a sentence or two** (in English) explaining its connection to mathematics. (provide description/translation if in Chinese)
- Do **not** simply give articles containing **numbers or data!**
- After collecting all 10 items, **write an additional passage** between **200 to 300 words** to comment on the mathematical contents and express your overall views about the items in relation to the **Learning Outcomes** of this course
- Submit **type-written** passage and 10 items in single **PDF** file (through Moodle).
- Due date: **16:55, 30th November, 2021.**

Examples

title



date



The New York Times: "Closing Gap Between Modern Life and Math Curriculum", 10th Feb 2013.

(<http://www.nytimes.com/2013/02/11/world/europe/11iht-educside11.html?ref=mathematics>)

source



clipping

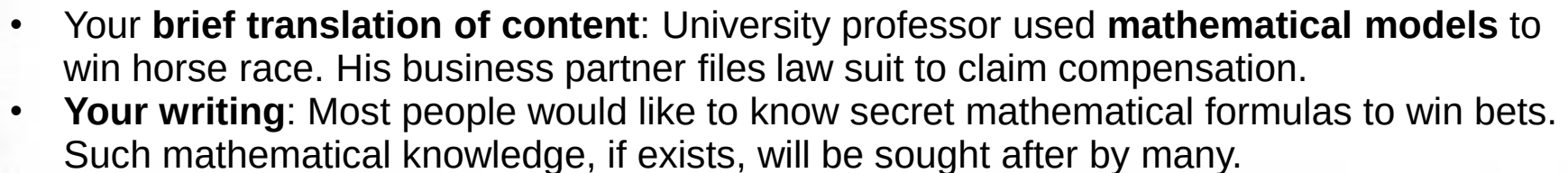


Conrad Wolfram, strategic director of Wolfram Research, is campaigning to change the mathematical curriculum in schools by making it more applicable to "real life". He argues that the mathematics education forced on children today is not at all useful in daily activities, and that modern life has become so quantitative (eg. through the use of statistics in medicine, comparing mortgages and understanding financial crises) we must completely change the mathematical curriculum so that we can deal with these new problems. He believes we should be retrained in computer programming, mental arithmetic and complex mathematics using technology.

Your writing



Example of raw material in Chinese (must summarize in English)



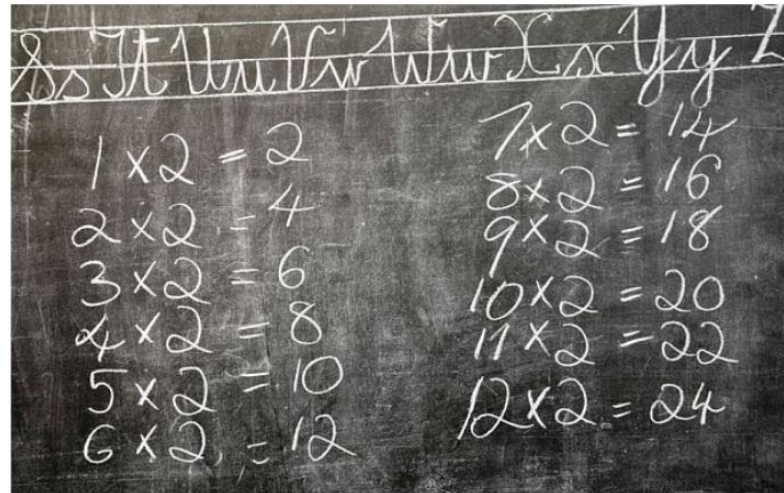
Examples

The Telegraph

<http://www.telegraph.co.uk/education/educationnews/11031288/Britains-maths-policy-simply-doesnt-add-up.html>

Britain's maths policy simply doesn't add up

As a woman wins the greatest prize, it's time to liberate girls from our test-driven culture



For the first time, the Fields Medal has been awarded to a woman – Maryam Mirzakhani, who is responsible for what fellow mathematicians call “probably the theorem of the decade” Photo: Alamy

By Jo Boaler

6:10AM BST 14 Aug 2014

Yesterday was an important day for gender equity in mathematics. For the first time, the Fields Medal was awarded to a woman – my colleague at Stanford University, Maryam Mirzakhani. It is a much-deserved prize for one of the most exciting mathematicians of our time.

Examples that are NOT acceptable

- Published on or before Sept. 14, 2021
- Not published in **mass media** with **general public** as audience
 - E.g., Journal articles like Nature, Bulletin of the American Mathematical Society, personal blogs, etc.
- Articles containing data but with **no analysis**.
 - Example: Bad: An article about the “Million People March against Police Brutality” – Description “One Million is a number.”
 - Better: An article about “how big” one million is; how long would it take to count to one million seconds?

Example: <http://nytimes.stats.com/nba/scoreboard.asp>

[HOME PAGE](#)
[TODAY'S PAPER](#)
[VIDEO](#)
[MOST POPULAR](#)
[TIMES TOPICS](#)

[Log In](#)
[Register Now](#)

Tuesday, January 19, 2016

Pro Basketball

[WORLD](#)
[U.S.](#)
[N.Y. / REGION](#)
[BUSINESS](#)
[TECHNOLOGY](#)
[SCIENCE](#)
[HEALTH](#)
[SPORTS](#)
[OPINION](#)
[ARTS](#)
[STYLE](#)
[TRAVEL](#)
[JOBS](#)
[REAL ESTATE](#)
[AUTOS](#)

[BASEBALL](#)
[PRO FOOTBALL](#)
[COLLEGE FOOTBALL](#)
[PRO BASKETBALL](#)
[COLLEGE BASKETBALL](#)
[HOCKEY](#)
[SOCCER](#)
[GOLF](#)
[TENNIS](#)
[GLOBAL SPORTS](#)

[Scoreboard](#)
[Stats](#)
[Standings](#)
[Teams](#)
[Players](#)
[Player News](#)
[Injuries](#)
[Transactions](#)

NBA Scoreboard

[Previous Day](#)
[Monday, January 18](#)
[Next Day](#)

Refresh: 60 Sec | 120 Sec | Off

Final-OT - 2 OT

1

2

3

4

OT

2OT

Tot

Philadelphia	29	18	16	33	11	6	113
New York	29	23	24	20	11	12	119

[Preview](#) | [Box](#) | [Recap](#)

Leaders

Philadelphia

New York

Points

J. Okafor 20

A. Afflalo 25

Rebounds

N. Noel 13

K. Porzingis 12

Assists

I. Smith 16

C. Anthony 7

Final

1

2

3

4

Tot

Portland	32	27	35	14	108
Washington	17	40	19	22	98

[Preview](#) | [Box](#) | [Recap](#)

Leaders

Portland

Washington

Points

C. McCollum 25

G. Temple 18

Rebounds

M. Plumlee 11

M. Gortat 13

Assists

M. Plumlee 7

J. Wall 10

Final-OT - 2 OT

1

2

3

4

OT

2OT

Tot

Utah	28	15	21	31	13	11	119
Charlotte	29	23	21	22	13	16	124

[Preview](#) | [Box](#) | [Recap](#)

Leaders

Utah

Charlotte

Points

G. Hayward 36

K. Walker 52

Rebounds

R. Gobert 14

2 tied with 9

Assists

G. Hayward 9

K. Walker 8

Final

1

2

3

4

Tot

Chicago	26	30	27	28	111
Detroit	32	21	26	22	101

[Preview](#) | [Box](#) | [Recap](#)

Leaders

Chicago

Detroit

Final

1

2

3

4

Tot

Orlando	17	21	21	22	81
Atlanta	24	27	21	26	98

[Preview](#) | [Box](#) | [Recap](#)

Leaders

Orlando

Atlanta

LATEST SCORES AND RESULTS

- [M.L.B.](#) | [N.B.A.](#) | [N.F.L.](#) | [N.H.L.](#)
- [NCAA Basketball: Men's](#) | [Women's](#)
- [Soccer: M.L.S.](#) | [E.P.L.](#) | [Champions League](#)
- [Golf: P.G.A.](#) | [L.P.G.A.](#) | [Web.com](#) | [Champions](#) | [European](#)
- [Tennis: Men](#) | [Women](#)
- [Motor Sports: NASCAR](#) | [Xfinity](#) | [Truck](#) | [IndyCar](#) | [Formula 1](#)

Group Essay / Presentation

1. Groups of size 2-3 (subject to enrollment sizes)
2. 2000 – 3000 words
3. On book(s) (or other media, e.g. movie) about mathematics that is course-relevant
4. If multiple books chosen, essay must relate them
5. Half-page plan due by **October 15, 2021, 16:55**
6. Oral presentations (short) in last 3 tutorials
7. Essay due by **December 3, 2021, 16:55.**

Group Project Logistics

1. Work in Groups of **3** (same tutorial group) – subject to confirmation of class size
2. 2000 – 3000 words
3. Half page essay plan: due **16:55, 15th October, 2021**
4. Oral Presentation:
 - File uploaded before **16:55, 8th Nov., 2021 (Mon.)**
 - During last 3 tutorial sessions
 - 12 Minutes plus 3 minutes Q&A
5. Essay: Due **16:55, 3rd December, 2021**

Group Project Logistics

6. On a book (or other media):
 - About Mathematics
 - Relevant to **Learning Outcomes**
7. Can use more than one (related) book.
8. Not a simple book report:
 - Your Interpretation
 - Connection to course/Learning Outcomes